MIL-C-46660 (Ord)
20 May 1962
SUPERSEDING
(See 6.5)

MILITARY SPECIFICATION

CARTRIDGE, 37MM, TP, M63 MOD. 1,
W/FUZE, BD, M58, PRACTICE,
LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling, and packing of one type of cartridge designated as Cartridge, TP, M63 Mod. I, for 37MM Subcaliber Gun.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-223 - Powder, Black.
MIL-A-2550 - Ammunition and Special Weapons, General Specification for.
MIL-I-45208 - Inspection Requirements, General Specification for.
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of.
MIL-P-46640 - Primer, Percussion M23A2 Loading, Assembling, and Packing.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Inspection Terms and Definitions.
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for percent Defective.

FSC: 1310

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DRAWINGS

ORDNANCE CORPS

8797461 - Box, Packing, Ammunition, for Cartridge, 37MM, TP, M63 Mod 1, for 37MM Subcaliber Guns in Fiber Container M127A2.
8831141 - Cartridge, TP, M63 Mode I, for 37MM Subcaliber Guns Assembly and Marking Diagram.
8833505 - Container, Fiber, Ammunition, M127A2 for Cartridge 37MM, TP, M63 Mod 1, for 37MM Subcaliber Guns.

PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes.
IEL-111856 - Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Cartridges.-The cartridges shall comply with all requirements specified on Drawing (dwg.) 8831141, and with all requirements specified in applicable specifications.

3.3 Propellant weight.-The weight of the propellant powder shall not differ by more than plus or minus 3 grains avoirdupois from that designated by the loading authorization for the particular lot of powder used.

3.4 Bullet pull.-The cartridge case shall separate from the projectile at a load of 500 pounds, minimum (min.) when tested as specified in 4.3.1.

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3.5 Proving ground.-The cartridge shall comply with the following requirements when tested as specified in 4.3.2.

3.5.1 Velocity.-The corrected velocity of the cartridge at ambient temperature shall be not more than 2130 feet per second nor less than 2070 feet per second and the standard deviation of the lot shall be not greater than 13 feet per second. (see 6.7)

3.5.2 Functioning.-The projectile shall produce a smoke cloud as a result of impact on the target.

3.5.3 Safety.-There shall be no hangfires or evidence of metal parts separation or breakup in the gunbore or in flight.

3.6 Moisture content of black powder.-The moisture content of the black powder used in loading the cartridges shall not exceed 1.25 percent at the time of loading. Testing shall be as specified in 4.3.4.

3.7 Workmanship.

3.7.1 Cartridge.-All manufacturing processing and assembly operations of the cartridge shall be correctly performed in a thoroughly workmanlike manner. Parts shall be of the proper finish (see Specification MIL-A-2550) and be dry and free of burrs, chips, dirt, grease, rust and other foreign material. The interior and exterior surfaces shall be coated as required by the applicable drawing. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

3.7.2 Loading of black powder.-The interior surface of the projectile shall be dry when the black powder is loaded into the projectile.

3.7.3 Threads.-The mating surfaces of the threads shall be free of black powder and all threads shall be free of foreign material. No liquid shall be used as a cleaner.

3.7.4 Exterior surface.-After loading, the exterior of the Cartridge shall be clean and free of black powder. The painting and marking shall be restored by the loading and assembling contractor on any component furnished to him painted or marked when such painting or marking has been marred or obliterated.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL-I-45208.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.
b. Number of units of product inspected.
c. Results obtained for all inspection performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of analysis on all material procured directly by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL-I-45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 Lot formation.-A lot shall consist of cartridges produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance, or fit, as determined by the Government, shall not necessitate changing the lot number. Each lot shall contain:

a. Cartridge cases from not more than one lot of resized cases, or

b. Cartridge cases of new manufacture of one interfix lot number from one manufacturer.

c. Projectiles of interfix lot number from one manufacturer.

d. Loaded practice fuzes of one interfix lot number from one manufacturer.

e. Loaded primers of one interfix lot number from one manufacturer.

f. Propellant from not more than one lot.

4.2.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

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4.2.2.1 Assembly, prior to loading propelling charge (see dwg. 8831141).

Categories	Defects	Method of inspection	Code Number. (see 6.6)
Critical:	None defined		
Major:	AQL 0.25 percent		
101.	Primer loose or damaged	Visual-Manual	01001
Minor:	AQL 0.40 percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	01002

4.2.2.2 Projectile prior to loading intermediate charge (see dwg. 8831142 covering a detail of dwg. 8831141).

Categories	Defects	Method of inspection	Code Number
Critical:	None defined		
Major:	AQL 0.25 percent		
101.	Weight of forward charge (prior to loading into projectile)	Gage	02001
102.	Distance from base of projectile to forward charge (after pressing)	Gage	02002
Minor:	None defined		

4.2.2.3 Projectile prior to loading upper charge (see dwg. 8831142 covering a detail of dwg. 8831141).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined		
Major:	AQL 0.25 percent		
101.	Weight of intermediate charge (prior to loading into projectile)	Gage	03001
102.	Distance from base of projectile to intermediate charge (after pressing)	Gage	03002
Minor:	None defined		

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4.2.2.4 Projectile prior to inserting cushion (see dwg. 8831142 covering a detail of dwg. 8831141).

Categories	Defects	Methods of inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Weight of upper charge (prior to loading into projectile)	Gage	04001
102.	Distance from base of projectile to upper charge (after pressing)	Gage	04002
103.	Contour of cavity in charge	Gage	04003
104.	Diameter of cavity in charge	Gage	04004
Minor: None defined			

4.2.2.5 Projectile prior to inserting fuze (see dwg. 8831142 covering a detail of dwg. 8831141).

Categories	Defects	Method of inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Cushion missing	Visual	05001
102.	Spacer missing	Visual	05002
103.	Gasket missing	Visual	05003
Minor:	AQL 0.40 percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	05004

4.2.2.6 Projectile (see dwg. 8831142 covering a detail of dwg. 8831141).

Cagegories	Defects	Method of inspection	Code No.
Critical: None defined			
Major:	AQL 0.25 percent		
101.	Fuze fails to withstand minimum disassembly torque	Test	06001
Minor:	AQL 0.40 percent		
201.	Total length, max	Gage	06002

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4.2.2.7 Assembly (see dwg. 8831141).

Categories	Defects	Method of inspection	Code No.
Critical:			
1.	Primer above flush	Gage	07001
Major:	AQL 0.25 percent		
101.	Cartridge case torn or cracked at mouth due to crimping	Visual	07002
Minor:	AQL 1.50 percent		
201.	Total length max	Gage	07003
202.	Protective coating with bare spot	Visual	07004
203.	Marking misleading or unidentifiable, including incorrect color code	Visual	07005
204.	Evidence of poor workmanship (see 3.7)	Visual	07006

4.2.2.8 Unsealed fiber container (see dwg. 8833505).

Categories	Defects	Method of inspection	Code No.
Critical: None defined			
Major:	AQL 1.50 percent		
101.	Packing component missing	Visual	08001
Minor:	AQL 1.50 percent		
201.	Cartridge cannot be removed by hand	Manual	08002
202.	Glue or asphalt on round	Visual	08003
203.	Cartridge improperly assembled in container	Visual	08004

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4.2.2.9 Sealed fiber container (see dwg. 8833505).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Inner layer of tape not in contact with container for at least 1/2 inch on each side of joint	Visual	09001
102.	Metal and loose or distorted	Visual-Manual	09002
103.	Container damaged through all impregnated asphalt layers, as evidenced by removing each layer of suspected container	Visual-Manual	09003
Minor:	AQL 2.50 percent		
201.	Tear tab length, min	Gage	09004
202.	Color of tape incorrect	Visual	09005
203.	Markings misleading or unidentifiable	Visual	09006
204.	Cuts, scuffs, or gouges in outer layer	Visual	09007
205.	Contents loose in container	Manual	09008
206.	Gap between body and cover of container (exceeding 1/8 inch)	Visual-Manual	09009

4.2.2.10 Sealed wooden packing box (see dwg. 8797461).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Hardware or strapping missing, broken or loose	Visual-Manual	10001
102.	Box damaged exposing contents	Visual	10002
103.	DOD symbol missing or incorrect	Visual	10003
104.	Board loose or broken	Visual-Manual	10004

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Minor:	AQL 1.50 percent		
201.	Car seal missing, unsealed or improperly attached	Visual	10005
202.	Markings misleading or unidentifiable	Visual	10006
203.	Handle missing or insecure	Visual-Manual	10007

4.2.3 Testing.

4.2.3.1 Propellant weight (see 3.3) Major defect Code No. 11001. The propellant charge shall be weighed and check-weighed 100 percent. The check-weighing shall be accomplished independently of the original weighing, using a different balance from that used to make the original weighing and, if performed manually, shall be performed by a different operator. Any propelling charge which fails to comply with the applicable requirement shall be classed defective and removed from the lot.

4.2.3.2 Bullet pull (see 3.4) Major defect, Code No. 07007.-The sampling plan for this test shall be in accordance with Standard MIL-STD-105, Table IVB, Code letter F and an AQL of 4.0 percent.

4.2.3.3 Profile and alignment (see dwg 8831141) Major defect Code No. 07008.-This test shall be conducted 100 percent. Any cartridge which fails to comply with the applicable requirement shall be classed defective and removed from the lot.

4.2.3.4 Proving ground.-The following tests shall be performed as specified in 4.3.2.

4.2.3.4.1 Velocity (see 3.5.1) Major defect Code No. 07009.

4.2.3.4.1.1 Beginning with the first lot produced and continuing until three consecutive lots shave complied with the acceptance criteria specified, the sampling plan for this test shall be in accordance with Standard MIL-STD-414 using Table B-3, Code letter G and an AQL of 1.00 percent. In addition, the lot shall be rejected if the product of the sample standard deviation and a factor of 0.69 exceeds the applicable lot requirement.

4.2.3.4.1.2 After three consecutive lots have met the criteria of 4.2.3.4.1.1, the sampling plan for this test shall be in accordance with Standard MIL-STD-414, using Table B-3, code letter D, and an AQL of 1.00 percent. In addition the lot shall be rejected if the product of the sample standard deviation and the factor 0.55 exceeds the applicable lot requirement.

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4.2.3.4.2 Functioning (see 3.5.2) Major defect Code No 070010

4.2.3.4.2.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, thirty cartridges shall be selected for this test plus the cartridges used in the velocity and safety tests. If more than three cartridges of the combined tests fail to meet the requirements the lot shall be rejected.

4.2.3.4.2.2 After three consecutive lots have met the criteria of 4.2.3.4.2.1, ten cartridges shall be selected for this test plus cartridges used in the velocity and safety test. If more than one cartridge of the combined tests fail to comply with the requirements the lot shall be rejected.

4.2.3.4.3 Safety (see 3.5.3) Critical defect Code No 07011

4.2.3.4.3.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, fifteen cartridges shall be selected for this test. If any cartridge in the velocity, functioning or safety fails to comply with the requirements the lot shall be rejected.

4.2.3.4.3.2 After three consecutive lots have met the criteria of 4.2.3.4.3.1, five cartridges shall be selected for this test. If any cartridge in the velocity functioning or safety fails to comply with the requirements the lot shall be rejected.

4.2.3.5 Moisture content of black powder, Major defect (see 3.6) Code No 12001.-A 5 gram sample of black powder shall be selected from each 8 hour working shift or part of for this test. If the sample fails to comply with the requirement of 3.6 the lot shall be rejected. The test shall be performed as specified in 4.3.4.

4.2.3.6 Check test for possible deterioration of Primer (see Specification MIL-P-46640 (Ord), Code No 13001.-If the total elapsed time between original acceptance of the M23A2 primer lot and the assembly of that lot into complete round cartridges exceeds two years or if the M23A2 primers have been subjected to adverse

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conditions, however brief, at any time since previous testing, the applicable lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in Specification MIL-P-46640 (Ord) immediately before the M23A2 primer lot is assembled into the cartridge. The test shall be performed by the contractor on assemblies selected by the Government inspector at the facility assembling the complete cartridge (see 6.2).

4.2.4 Inspection equipment Index of Inspection Equipment List Number 111856 identifies avaiable approved inspection equipment designs. The contractor may, however, design alternate inspection equipment in order to facilitate his operation. Designs of any acceptance inspection equipment prepared by the contractor to support his approved written inspection plan (see 4.1.1) shall be approved by the Government prior to fabrication or procuring of such equipment (see 6.4). All acceptance inspection equipment shall be certified by the Government as complying with the approved design prior to its use. Supply and maintenance of inspection equipment shall be in accordance with Specification MIL-I-45607.

4.3 Test methods and procedures.

4.3.1 Bullet pull.-The sample cartridges shall be tested in an appropriate tensile testing machine to determine the security of the crimp of the cartridge case to the projectile by applying a load in the axial direction; rate of travel of the testing machine head shall be from 0.100 to 0.125 inches per minute. The individual readings shall be recorded and reported.

4.3.2 Proving ground.

4.3.2.1 Velocity and functioning.-The tests shall be conducted at a government proving ground in accordance with the applicable ordnance Proof Manual and Acceptance Test Procedures. The target shall be a six inch thick wood bursting screen placed 400 feet from the muzzle of the gun. The sample cartridges shall be fired in a gun for which the cartridge is standard and observation made for conformance to all applicable requirements.

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4.3.2.2 Safety.-The rounds shall be assembled with a charge adjusted to give a pressure of 112 percent of the average pressure for the particular lot of propellant. The test shall be conducted as specified in 4.3.2 and observed for conformance with the applicable requirement.

4.3.3 Moisture content of black powder.-The moisture content of the black powder shall be determined in accordance with Specification MIL-P-223.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A.-The cartridges shall be preserved and packaged in accordance with dwg. 883505.

5.2 Packing

5.2.1 Level A.-The cartridges shall be packed in accordance with dwg. 8797461.

5.3 Data cards.-Data card shall be prepared for each lot in accordance with the information in Specification MIL-A-2550.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.2 Cost of check tests.-The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of check tests. The test shall be conducted at Government expense without cost to the contractor who loaded the primers or to the contractor assembling the cartridge, and shall not constitute a basis for rejection against either contractor, except where deterioration has occured as a direct result of carelessness in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

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6.3 Submission of test data.-In addition to the normal distribution of records for each lot of cartridges, when the item is procured by the Department of the Army, copies shall be sent to:

1 copy - Commanding Officer
Picatinny Arsenal
Attn: ORDBB-ND4
Dover, New Jersey

6.4 Submission of contractor developed inspection equipment designs.-Documentation of contractor developed inspection equipment design indentified with the associated examination or test required by the specification shall be submitted in duplicate to Picatinny Arsenal, Dover, New Jersey, Attention ORDBB-ND, for approval. This documentation shall include special equipment designs and a complete description of the use and application of all standard devices and set-ups as well as the special equipment designs. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment design is permissible. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.5 Supersession data.-This specification supersedes in part MIL-S-1393, dated 16 September 1949, covering Shell, Fixed HE, M63 for 37MM Guns TP, M63 Mod I for 37MM Guns TP, M92 for 37MM Guns Assembling, Marking, and Packing.

6.6 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.7 Velocity.-The velocity limits are tentative because of a lack of recent firing data. Therefore, limits are subject to change based on performance capability demonstrated in rounds of new manufacture.

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Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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